STUDYLAB HUB · AP & COLLEGE BIOLOGY · UNIT 2 (CHAPTERS 6–7)

Unit 2: Cell Structure & Membrane Transport

Organelles · Surface-Area-to-Volume Ratio · Fluid Mosaic Model · Osmosis & Water Potential

AP & College Biology — Unit 2 (Chapters 6–7): Cell Structure, Membranes & Transport

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1. Unit 2 Roadmap (Chapters 6–7)

Topic	Core Biological Principle	What Exams Actually Test
2.1–2.2 · Subcellular Components	Compartmentalization separates incompatible metabolic reactions and increases internal membrane surface area	Tracing a secreted protein from synthesis to exocytosis: **Rough ER → Transport Vesicle → cis-Golgi → trans-Golgi → Plasma Membrane**
2.3 · Cell Size (SA : V Ratio)	Surface Area-to-Volume Ratio (SA/V) decreases as radius increases (for a sphere, SA/V = 3 ÷ r)	Calculating SA/V for cubes/spheres and explaining why small cells (or microvilli/cristae folds) exchange nutrients and wastes faster
2.4–2.5 · Plasma Membrane & Permeability	Fluid mosaic of amphipathic phospholipids, cholesterol (fluidity buffer), and integral/peripheral proteins	Predicting what crosses the hydrophobic core unaided (**small, nonpolar molecules**: O_2, CO_2, N_2, steroids) vs. what requires transport proteins (**ions & large polar molecules**: Na^+, K^+, H^+, glucose, H_2O via aquaporins)
2.6–2.7 · Passive vs. Active Transport	Passive transport moves solutes **down** their gradient ($\Delta G < 0$, no ATP); Active transport pumps solutes **against** their gradient (requires ATP or cotransport)	Distinguishing simple diffusion, facilitated diffusion (channels/carriers), primary active transport (Na^+/K^+ ATPase, proton pumps), and secondary active cotransport
2.8 · Tonicity & Water Potential (Ψ)	Water moves from **higher water potential (Ψ) / hypotonic** (lower solute concentration) to **lower water potential (Ψ) / hypertonic** (higher solute concentration)	Calculating solute potential **$\Psi s = -iCRT$** and total water potential **$\Psi = \Psi p + \Psi s$**, and predicting plasmolysis vs. turgor vs. cytolysis
2.10–2.11 · Endosymbiosis & Compartmentalization	Mitochondria and chloroplasts evolved from engulfed aerobic and photosynthetic prokaryotes	Citing the **4 structural lines of evidence** for endosymbiosis (circular DNA, 70S ribosomes, double membranes, binary fission)

2. The Endomembrane Protein Secretion Pathway (Chapter 6)

When a pancreatic beta cell synthesizes and secretes the protein hormone **insulin**, follow the exact organelle sequence:

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Ribosome on Rough ER (translates mRNA into polypeptide inside ER lumen)
   → Rough Endoplasmic Reticulum (folds protein, adds carbohydrate tag / glycosylation)
   → Transport Vesicle buds from transitional ER
   → Golgi Apparatus (cis face → medial cisternae → trans face: modifies, sorts & packages)
   → Secretory Vesicle buds from trans-Golgi
   → Fuses with Plasma Membrane (Exocytosis releases protein outside cell)
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> **Exam Trap**: Free cytosolic ribosomes synthesize proteins that stay **inside the cytosol** (e.g., glycolysis enzymes) or are imported into the nucleus, mitochondria, or chloroplasts. Bound ribosomes on the **Rough ER** synthesize **secreted proteins, membrane proteins, and lysosomal enzymes**.

3. Organelle Structure-to-Function Master Table

Organelle	Structural Feature	Exact Metabolic Function
Nucleus & Nucleolus	Double nuclear envelope with nuclear pores; dense nucleolus region	Houses chromatin (DNA + histones); nucleolus transcribes rRNA and assembles ribosomal subunits
Rough ER	Network of membranes studded with ribosomes	Synthesizes secreted/membrane proteins; performs initial N-linked glycosylation and folding
Smooth ER	Tubular membrane network lacking ribosomes	Synthesizes lipids/steroid hormones, detoxifies drugs/poisons (liver), and stores **Ca^{2+} ions** (sarcoplasmic reticulum in muscle)
Golgi Apparatus	Flattened membrane sacs (cisternae) with *cis* (receiving) and *trans* (shipping) faces	Modifies glycoproteins/glycolipids, sorts cargo, and packages vesicles for secretion or lysosomes
Lysosome	Acidic vesicle (pH ≈ 4.5–5.0) containing hydrolytic enzymes	Intracellular digestion (autophagy of damaged organelles and phagocytosis of pathogens); compartmentalization prevents self-digestion of the neutral cytosol (pH ≈ 7.2)
Mitochondrion	Smooth outer membrane + highly folded inner membrane (**cristae**) enclosing the **matrix**	Cristae folds maximize surface area for the electron transport chain and ATP synthase (oxidative phosphorylation)
Chloroplast	Double outer envelope + stacked **thylakoid** membranes (**grana**) suspended in **stroma**	Thylakoid membranes capture light and generate ATP/NADPH; stroma runs the Calvin cycle to fix CO_2 into sugar

4. Water Potential (Ψ) & Osmosis Formula Sheet (Chapter 7)

1. **Total Water Potential Equation**:

- **Ψ = Ψp + Ψs**
- **Ψp** = Pressure potential (for an open beaker, **Ψp = 0.0 bars**; inside a turgid plant cell, Ψp > 0).
- **Ψs** = Solute potential (osmotic potential), which is always **0 or negative** (adding solute lowers Ψs).

2. **Solute Potential Equation**:

- **Ψs = −i · C · R · T**
- **i** = Ionization constant (number of particles per formula unit in water: **i = 1** for sucrose or glucose; **i = 2** for NaCl → Na^+ + Cl^-; **i = 3** for $CaCl_2$ → Ca^{2+} + 2 Cl^-).
- **C** = Molar concentration (mol/L or M).
- **R** = Pressure constant = **0.0831 liter·bars / (mol·K)**.
- **T** = Absolute temperature in Kelvin (**T = °C + 273**).

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5. Worked Exam Practice (MCQs + Scored FRQ)

Worked Calculation Problem

A plant root cell has an internal solute potential of Ψs = −7.5 bars and a turgor pressure potential of Ψp = +2.5 bars. It is placed into an open beaker containing a 0.30 M sucrose solution at 27°C. Calculate the water potential of the beaker solution and state the direction of net water movement.

- **Step 1 (Cell Ψ)**: Ψ(cell) = Ψp + Ψs = +2.5 + (−7.5) = **−5.0 bars**.
- **Step 2 (Beaker Ψs)**: For sucrose, i = 1, C = 0.30 M, R = 0.0831 L·bar/(mol·K), T = 27 + 273 = 300 K:

Ψs(beaker) = −(1)(0.30)(0.0831)(300) = **−7.48 bars**.

- **Step 3 (Beaker Ψ)**: Because the beaker is open to the atmosphere, Ψp = 0, so **Ψ(beaker) = −7.48 bars**.
- **Step 4 (Direction of Water Flow)**: Water always flows from **higher (less negative) Ψ to lower (more negative) Ψ**. Since −5.0 bars > −7.48 bars, net water flows **out of the plant cell into the hypertonic beaker solution**, causing **plasmolysis** (turgor pressure drops).